Forecasting Retail Demand for Production and Logistics Planning

A Time Series Case Study Using ***Walmart Sales Data***

Description: In this project, we plan to analyze and forecast retail sales using the Walmart Sales Forecasting dataset from Kaggle, which contains weekly sales data across multiple stores and departments. The dataset includes factors such as holidays, store characteristics, and seasonal shopping patterns, making it a realistic foundation for production and logistics planning.

We would begin with preparing and exploring the time series to identify underlying trends, recurring seasonal patterns, and any anomalies that may affect forecasting accuracy. For the modeling stage, we use two approaches: a simple Holt-Winters (Holt's) method without seasonal components to capture basic level and trend, and an advanced forecasting approach using Facebook Prophet, which can flexibly model seasonality, holiday effects, and trend changes. After comparing the models using standard evaluation metrics, we translate the forecasting results into practical implications for production scheduling, inventory management, capacity utilization, and transportation planning.